Jacob Karr is a Creative Programmer.

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EXPERIENCE

UNITY CODING TEACHER @ CONCORDE EDUCATION

May '23 - Present

- Plans and facilitates collaborative learning sessions for students of diverse backgrounds, academic profiles, and skill levels.
- Assists students in the development of computer science and coding skills, adapting teaching methods to better meet student needs.
- Maintains consistent communication with the Director of Educational Development to ensure alignment with program goals.

UNITY AR ENGINEER (CONTRACT)

May '21 - July '22

- Spearheaded the development and release of *Stout's Mobile Augmented Reality Tour of Art (SMART-Art)* using the Unity 3D engine, leading a team through the entire software development lifecycle.
- Collaborated with the product owner to determine the most effective approach for meeting project requirements, holding frequent meetings to clarify design and receive feedback.
- Streamlined the development pipeline by creating in-engine tools, resulting in increased efficiency and a more user-friendly workflow.
- Oversaw user testing to ensure compliance with AR and UI/UX design principles, resolving issues and implementing changes as necessary.
- Authored comprehensive technical documentation and user guides to facilitate future development and ensure the continuity of the project.
- Integrated the augmented reality framework into a digital humanities course as a teacher's assistant, developing engaging lessons and activities that leveraged the technology.

GAME PROGRAMMER AND TECHNICAL ARTIST (CAPSTONE)

Sept. '20 - May '21

- Worked closely with a diverse team of programmers, artists, and musicians, fostering collaboration and communication across disciplines.
- Designed and implemented custom shaders using HLSL and Unity's shader graph, enhancing the overall quality of the project.
- Supported lighting development and collaborated with artists to ensure a cohesive visual style, providing technical expertise and assistance as needed.
- Responsible for code review: spot and refactor code smell, improve performance, and verify proper documentation and code style.

EDUCATION

THE UNIVERSITY OF WISCONSIN-STOUT

B.S. Computer Science: Game Design & Development, Minor in Mathematics

Graduated in May '22

SKILLS

Unity, Unreal
C#, C, C++, Java, Python
GLSL, HLSL
HTML, CSS, JavaScript, Phaser, Three.js
Git, Windows Terminal
Scrum, Kanban, Technical Writing, Knowledge Graphs

EXTRAS

Awarded 'Best Mobile Game' at Stout Game Expo 2018
Prolific game jam attendee